Buyside Quant Prep Roadmap

NewWhale Career

Each module mapped to the decks that cover it

1. Probability and Brainteasers

Q03-Q08, Q12, Q13

Classical probability, combinatorics (dice, coins, cards), Bayes, expectation and variance, Markov chains and gambler's ruin, uniform / geometric / normal distributions, symmetry arguments.

Every quant seat. Market makers test this live and fast.

2. Statistics: Estimation and Inference

Q10 (survey), Q21 (deep)

Bias / variance / MSE of estimators, method of moments, MLE and its properties, laws of total expectation and variance, confidence intervals, hypothesis testing, min / max of random variables.

HF research seats; the warm-up layer before regression questions.

3. Statistics: Regression and Model Building

Q19, Q20 (the core pair)

OLS from objective to normal equations, all six assumptions and their failure modes, t / F inference, R-squared traps, multicollinearity and VIF, bias-variance, cross-validation (incl. walk-forward), gradient descent, ridge / lasso / elastic net.

The highest-yield technical area for hedge fund and buyside interviews.

4. Options Theory

S09-S10 (foundations), Q22 (Greeks), Q09 (stochastic calculus)

Payoffs and put-call parity, Black-Scholes formula and assumptions, implied volatility and the smile, and how every Greek behaves deep OTM / ATM / deep ITM and as expiry approaches.

Options desks and vol funds; Q09 only for researcher / pricing seats.

5. Programming

Q14, plus daily practice outside the slides

Python first (C++ later, only after Python is solid). OOP, lists / tuples / dicts / heaps, hashmaps, big-O, searching and sorting, pandas data manipulation, mutable vs immutable. Practice easy-to-medium algorithm problems until 20-minute solves feel routine.

Screens and pair-coding rounds at every systematic shop.

6. Applied Projects

QM series (hands-on modeling track)

Build one real, validated strategy you can defend for 20 minutes. Project menu: market-beta (CAPM) study, GARCH volatility forecast, pairs trade with cointegration test, gradient-boosted return classifier, factor-model replication.

The differentiator: interviews end up spending most time here.

7. Question Bank and Mocks

Q16 (easy), Q17 (medium), Q18 (hard), Q15 (fit), Q02 (mental math)

112 re-authored interview problems with full solutions. Work them on paper before reading answers. Finish with timed mock interviews.

The last two weeks of prep should mostly live here.

The 8-Week Plan

NewWhale Career

A realistic schedule at 5 to 8 focused hours per week; compress to 4 weeks only if full-time

Weeks 1-2	Probability core: Q03 (fundamentals), Q04 (expected value), Q05 (counting), Q06 (Bayes). Ten easy bank questions per week. Five minutes of mental math (Q02) daily.
Week 3	Distributions and processes: Q07, Q08. Classic puzzles Q13. Keep the easy bank running.
Week 4	Statistics I: Q10 as the map, then Q21 in depth. Derive the Bernoulli and Normal MLEs on paper without notes.
Week 5	Statistics II: Q19 then Q20, the highest-yield week of the plan. Reproduce the OLS derivation and the ridge / lasso contrast on paper.
Week 6	Options: S09 / S10 if new to options, then Q22 until you can rebuild both Greek tables unprompted. Researcher-track candidates add Q09.
Week 7	Programming: Q14 patterns, daily easy / medium problems in Python, one pandas exercise per day. Start the medium bank (Q17).
Week 8	Integration: finish Q17 / Q18, one applied project write-up (QM track), Q15 behavioral prep, and at least one timed mock.

What each seat tests

Market makers and prop shops

Mental math under time pressure, probability and EV games, market-making intuition (Q11), game theory (Q12). Speed and composure matter as much as correctness.

Hedge funds (fundamental or pod)

Regression and model building (Q19 / Q20), statistics, a defensible project, and market awareness. Expect 'here is a dataset, how would you find a signal?'

Quant research at systematic funds

Everything above plus stochastic calculus (Q09) and deeper ML discussion. Your project write-up carries the most weight here.

Rule of thumb: an interviewer can take any answer one level deeper. Prep one level deeper than the answers you plan to give.